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← MAT 285 Intro to Diff Eq, section H001, Spring 2026

HW 2.3 (Homework)

INSTRUCTOR
Jacob Chapman
University of Southern Mississippi

Due Date: THU, FEB 12, 2026 11:59 PM CST

REQUEST EXTENSION

Current Score: – / 39 POINTS | 0.0 %

Scoring and Assignment Information

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13
POINTS	– / 3	– / 3	– / 3	– / 3	– / 3	– / 3	– / 4	– / 5	– / 2	– / 2	– / 3	– / 2	– / 3

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

1. **[– / 3 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.001.

Find the general solution of the given differential equation.

$$\frac{dy}{dx} = 8y$$

$y(x) =$

Give the largest interval over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution. (Enter the transient terms as a comma-separated list; if there are none, enter NONE.)

Resources

Read It

2. **[– / 3 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.003.

Find the general solution of the given differential equation.

$$\frac{dy}{dx} + y = e^{2x}$$

$y(x) =$

Give the largest interval over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution. (Enter the transient terms as a comma-separated list; if there are none, enter NONE.)

Resources

Read It Watch It

3. **[– / 3 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.004.

Find the general solution of the given differential equation.

$$4\frac{dy}{dx} + 20y = 5$$

$y(x) =$

Give the largest interval I over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution.

Resources

Read It

4. **[– / 3 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.005.

Find the general solution of the given differential equation.

$$y' + 3x^2y = x^2$$

$y(x) =$

Give the largest interval over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution. (Enter the transient terms as a comma-separated list; if there are none, enter NONE.)

Resources

Read It

5. **[– / 3 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.007.

Find the general solution of the given differential equation.

$$x^2y' + xy = 9$$

$y(x) =$

Give the largest interval over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution. (Enter the transient terms as a comma-separated list; if there are none, enter NONE.)

Resources

Read It Watch It

6. **[– / 3 Points]** DETAILS MY NOTES ASK YOUR TEACHER

ZillDiffEQModAp11 2.3.009.

Find the general solution of the given differential equation.

$$x\frac{dy}{dx} - y = x^2 \sin(x)$$

$y(x) =$

Give the largest interval over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution. (Enter the transient terms as a comma-separated list; if there are none, enter NONE.)

Resources

Read It

7. **[– / 4 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.028.EP.

Consider the following differential equations.

$$y\frac{dx}{dy} - x = 6y^2, \quad x(9) = 1$$

Find the coefficient function $P(y)$ when the given differential equation is written in the standard form $\frac{dx}{dy} + P(y)x = f(y)$.

$P(y) =$

Find the integrating factor for the differential equation.

$e^{\int P(y)dy} =$

Solve the given initial-value problem.

$x(y) =$

Give the largest interval I over which the solution is defined. (Enter your answer using interval notation.)

$I =$

Resources

Read It Watch It

8. **[– / 5 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.038.EP.

Consider the following differential equation.

$$\frac{dy}{dx} + y = f(x),\ y(0) = 1, \text{ where } f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ -1, & x > 1 \end{cases}$$

Find the coefficient function $P(x)$ when the given differential equation is written in the standard form $\frac{dy}{dx} + P(x)y = f(x)$.

$P(x) =$

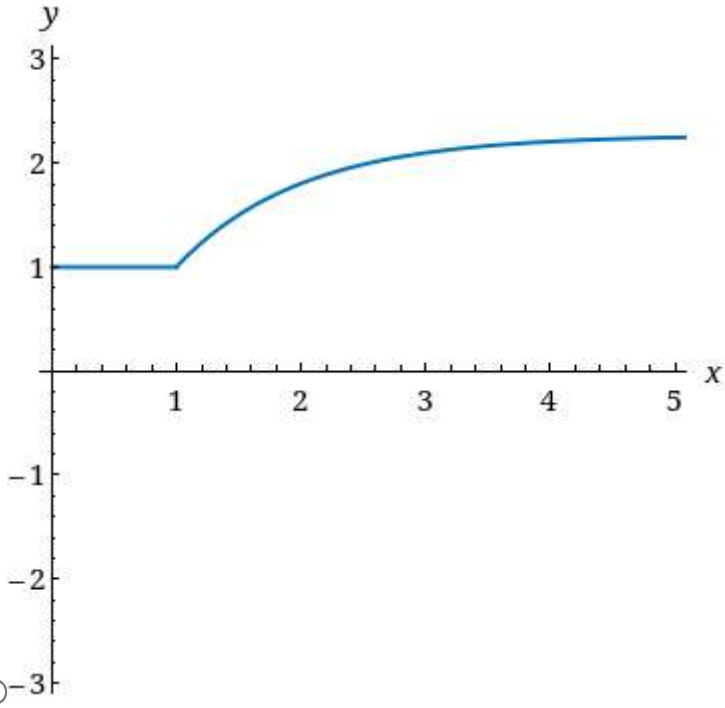
Find the integrating factor for the differential equation.

$e^{\int P(x)\,dx} =$

Proceed as in Example 6 in Section 2.3 to solve the given initial-value problem.

$$y(x) = \begin{cases} ____, & 0 \le x \le 1 \\ ____, & x > 1 \end{cases}$$

Use a graphing utility to graph the continuous function $y(x)$.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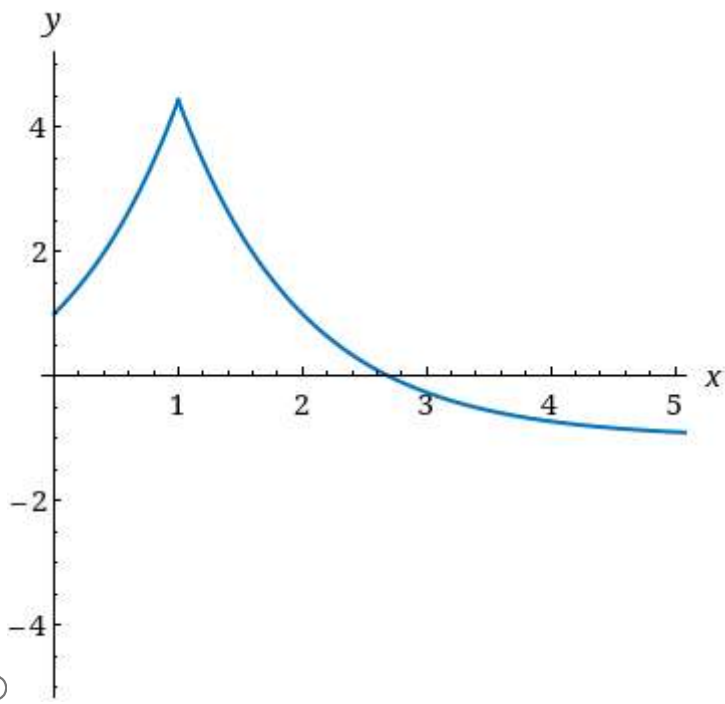
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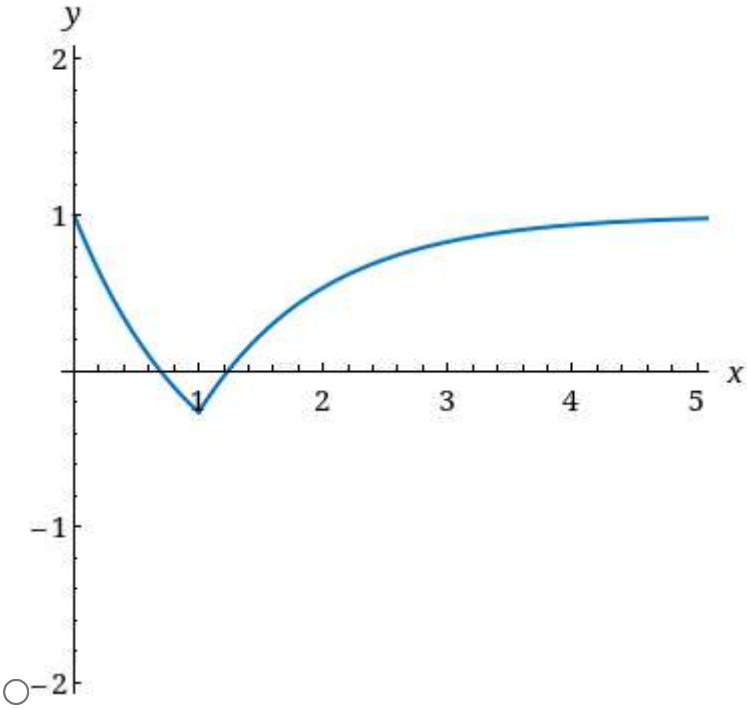

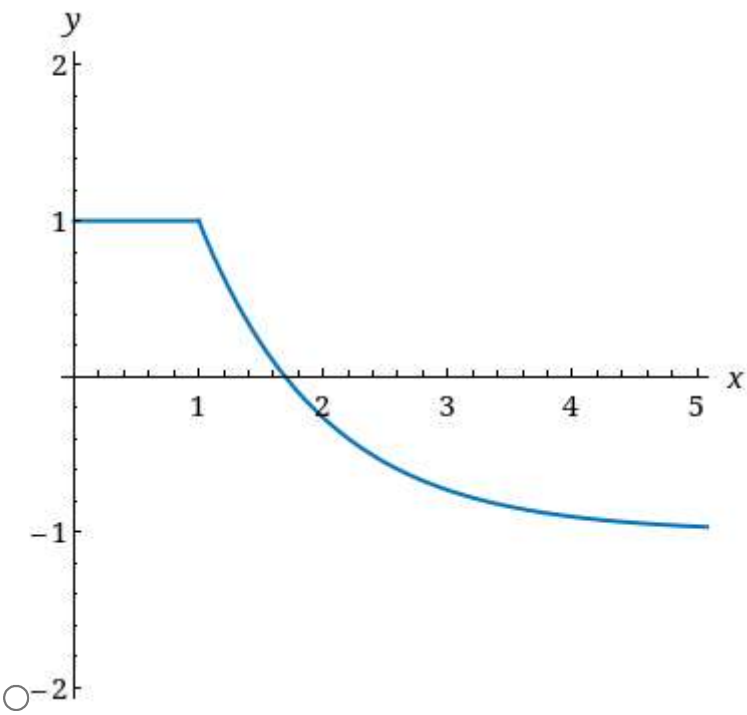

Resources

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9. **[– / 2 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.031.

Solve the given initial-value problem.

$$x\frac{dy}{dx} + y = 2x + 1, \quad y(1) = 9$$

$y(x) =$

Give the largest interval I over which the solution is defined. (Enter your answer using interval notation.)

$I =$

Resources

Read It Watch It

10. **[– / 2 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER

ZillDiffEQModAp11 2.3.034.

Solve the given initial-value problem.

$$x(x+1)\frac{dy}{dx} + xy = 1, \quad y(e) = 1$$

$y(x) =$

Give the largest interval I over which the solution is defined. (Enter your answer using interval notation.)

$I =$

Resources

Read It

11. **[– / 3 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.022.

Find the general solution of the given differential equation.

$$\frac{dP}{dt} + 2tP = P + 6t - 3$$

$P(t) =$

Give the largest interval over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution. (Enter the transient terms as a comma-separated list; if there are none, enter NONE.)

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12. **[– / 2 Points]**

DETAILS | MY NOTES | ASK YOUR TEACHER | PRACTICE ANOTHER

ZillDiffEQModAp11 2.3.025.MI.

Solve the given initial-value problem.

$$\frac{dy}{dx} = x + 2y, \quad y(0) = 7$$

$y(x) =$

Give the largest interval I over which the solution is defined. (Enter your answer using interval notation.)

$I =$

Resources

Read It Watch It

13. **[– / 3 Points]** DETAILS MY NOTES ASK YOUR TEACHER

ZillDiffEQModAp11 2.3.013.

Find the general solution of the given differential equation.

$$x^2y' + x(x + 2)y = e^x$$

$y(x) =$

Give the largest interval over which the general solution is defined. (Think about the implications of any singular points. Enter your answer using interval notation.)

Determine whether there are any transient terms in the general solution. (Enter the transient terms as a comma-separated list; if there are none, enter NONE.)

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